

BORA VAMSI REDDY

Embedded Systems Engineer (Fresher)

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EDUCATION

Degree	Specialization	Institute	Year	CGPA
B. Tech	Electronics & Comm. Engineering	NSRIT	2022-2025	8.42
Diploma	Electronics & Comm. Engineering	Govt. Chodavaram	2019-2022	8.0

INTERSHIPS and TRAINING

- **Trainee** [Vector India Institute, Hyderabad] June 2025 - January 2026
 - Gained hands on training in **Embedded C**, **ARM** Core architecture, Linux fundamentals, and Serial Communication **Protocols (I2C, UART, SPI and CAN 2.0)**.
 - Worked on low-level programming concepts including memory, pointers, and peripheral interfacing.
- **Intern** [BSNL – Bharat Sanchar Nigam Limited] May-Nov 2024
 - Finished 45 days of training in **telecommunications**, gaining exposure to mobile communication networks and understanding **ISP infrastructure** and operations.
 - Obtained 4 months of hands-on experience in **optical fiber networks**, including tower installations, server maintenance, and assisting in **fiber splicing** and **fusion**.
- **Summer Intern** [HMI Engineering Services] June-July 2023
 - **30 days** Trainee in **Embedded Systems & IoT**, worked on **beauty of Embedded Systems** i.e., **Sensors** and **Actuators** (Ultrasonic Sensor, IR Sensor etc).

PROJECTS

- **Attendify-RFID: Smart Employee Tracking & Reporting System**
 - Implemented an RFID-based attendance system using the **LPC2148 (ARM7) microcontroller**, with an **SPI-based EEPROM** to store admin authorization IDs locally.
 - Established **UART1 communication with the RFID reader** and **UART0 for PC-side file-based employee data logging**, enabling secure authentication and centralized record management.
- **PowerChrono: The Future of Time-Driven Energy Automation**
 - Built a real-time, schedule-based device control system on the **LPC2148 (ARM7)** using its **internal RTC** to switch the device ON and OFF automatically at configured times.
 - Implemented a **clean, menu-driven interface** using **two external interrupts for time editing and schedule viewing**, along with a **high-priority FIQ interrupt** that safely exits any menu and returns the system to a defined **home screen**.
- **Audio and Data Transfer using Light Fidelity (Li-Fi)** [Asst. Prof. Ramanaji Kalla]
 - Developed a project on **Audio and Data Transfer** using Light Fidelity (**Li-Fi**), leveraging visible light communication for **high-speed, secure** data transmission.
 - Designed and executed a system to encode, transmit, and decode multimedia data. Enhanced knowledge in **optical communication**, **embedded systems**, and **signal processing**.
- **I2C Protocol using Edge Artix 7 FPGA** [Prof. Dr. Rajashekhar]
 - Engineered a fully functional **I2C master** controller module in **Verilog**, adhering to the I2C protocol specifications and implemented essential I2C functionalities, including **start/stop** conditions, address generation, **read/write operations**, and acknowledge/not-acknowledge handling.
 - Improved my understanding of **serial communication protocols**, and embedded system design.
- **SPI DAC using Edge Artix 7 FPGA** [Prof. Dr. Rajashekhar]
 - Developed a custom **Verilog** module to handle the **SPI protocol**, including clock generation, data serialization, and chip select management, ensuring accurate data transmission to the DAC.
 - This project provided hands-on experience in debugging and troubleshooting Verilog modules, enhancing problem-solving skills.

TECHNICAL SKILLS

- **Programming:** C, Embedded C, experience with 8051 programming.
- **Microcontrollers:** LPC2129 / LPC2148 (ARM7), 8051 family (AT89C51), Arduino microcontrollers.
- **Communication Protocols:** UART, I²C, SPI; CAN (theoretical understanding).
- **Interrupts:** In-depth knowledge of LPC2148 interrupt architecture, including IRQ and FIQ handling.
- **Hardware Interfacing:** LCDs, ADC, DAC, EEPROM, RFID readers, and peripheral devices.
- **Development Tools & Simulators:** Keil µVision, Proteus, Wokwi, Tinkercad.

SOFT SKILLS

- Detail-oriented approach to debugging and system design
- Ability to understand low-level concepts and apply them practically
- Patience and persistence in troubleshooting complex issues
- Ability to work independently as well as in a team environment.

CERTIFICATIONS

- **NPTEL** - Embedded Systems Design (ESD) with **75%**.
- Internship on **FPGA** by Pantech Solutions (30 days). _____
- **NPTEL** – An Introduction to Internet of Things (IoT) with **79%**. _____